



ScriptureCast

Product Requirements Document (PRD) v1.0

 [Date] • Status: Draft

1. Executive Summary

Problem: When pastors call out Bible verses during sermons, congregants scramble to find them in apps or Bibles, disrupting focus. Verses displayed manually by a technician are delayed, error-prone, and distract from the sermon flow.

Solution: An AI-powered system that listens to the pastor's live audio, instantly detects verse references, and displays the verse text on screens — with no human intervention required.

Value Proposition: Congregants stay engaged, verses appear within 2 seconds of being spoken, and pastors preach without technical distractions.

2. User Personas

Persona	Needs	Pain Points
Pastor (primary)	Say verse → appears on screen. Cancel button if wrong. No training.	Current system: must wait for tech, or no display at all
Technical Director	Monitor health, override, adjust sensitivity	No maintenance burden during service
Congregant	See verse clearly from any seat, enough time to read	Can't find verses in their own Bible fast enough
Church Planter (buyer)	Affordable (\$500–1500), offline, simple hardware	Complex AV setups, recurring fees

3. Functional Requirements

3.1 Audio Capture & Processing

ID	Requirement	Priority
AC-01	Accept live audio from standard XLR or USB microphone	P0
AC-02	Filter background music, congregation noise, reverb	P1
AC-03	<500ms latency from mic to transcription engine	P0
AC-04	Work offline (no internet) using local ASR model	P1

3.2 Verse Detection

ID	Requirement	Priority
VD-01	Detect verse references in natural speech ("John 3:16", "Romans 8, verse 28")	P0
VD-02	Recognize 66 books + common abbreviations (Gen, Jn, Rev, Psa)	P0
VD-03	Chapter-only references ("Turn to Genesis 1") → display first verse	P2
VD-04	Support verse ranges ("John 3:16-21")	P1
VD-05	Confidence threshold: 85% auto-display; 60–84% requires confirmation	P1

3.3 Verse Display

ID	Requirement	Priority
DS-01	Show reference in bold, 20% of screen height	P0
DS-02	Verse text minimum 72pt font (readable from 100ft)	P0
DS-03	Outputs: HDMI, NDI stream, web page URL, ProPresenter	P1
DS-04	Configurable duration (default 15 sec, 5–60 sec)	P0
DS-05	Pastor clears via foot pedal, wireless button, or mobile app	P0

3.4 Bible Data & Translations

ID	Requirement	Priority
BD-01	Include KJV + World English Bible (public domain)	P0
BD-02	Additional translations (ESV, NIV, NLT) – licensing required	P2
BD-03	Pastor can set default translation per service	P1
BD-04	Cache entire Bible locally for offline access	P0

4. Non-Functional Requirements

Category	Requirement	Target
Latency	Audio → verse on screen	≤2 seconds (80th percentile), ≤3 sec max
Precision	Verse detection false positives	> 95% (no false positives)
Recall	Catches intended verses	> 90%
Availability	Uptime during service	99.9%
Offline	Full functionality without internet	Yes
Hardware	Minimum CPU	Intel i5 / Raspberry Pi 4 (8GB)

5. Success Metrics (KPIs)

≤3 sec

Detection latency (p95)

≤1

False positive per hour

≤2

Missed verses per service

4/5

Pastor satisfaction

≤3

Operator interventions per service

6. Risks & Mitigations

Risk	Mitigation
Pastor speaks too fast / unclear	Training mode + adjustable sensitivity
Background music triggers false detection	Noise gate + bandpass filter (300Hz–3kHz focus on speech)
Offline ASR too slow on Raspberry Pi	Use smaller Whisper model (tiny.en); pre-warm model
Book name variations fail (e.g., "Song of Solomon")	Expand normalization table to 200+ variants

7. Phased Roadmap



Phase 1: MVP (2–3 months)

- Local microphone input • Offline ASR (Whisper.cpp) • 10 most common books + verse parsing
- KJV + WEB translations • HDMI output + simple web display • Wireless clear button



Phase 2: Polished (1–2 months after MVP)

- 66 books + all abbreviations • Operator dashboard • Confidence-based confirmation mode
- ProPresenter integration • Verse ranges



Phase 3: Advanced (future)

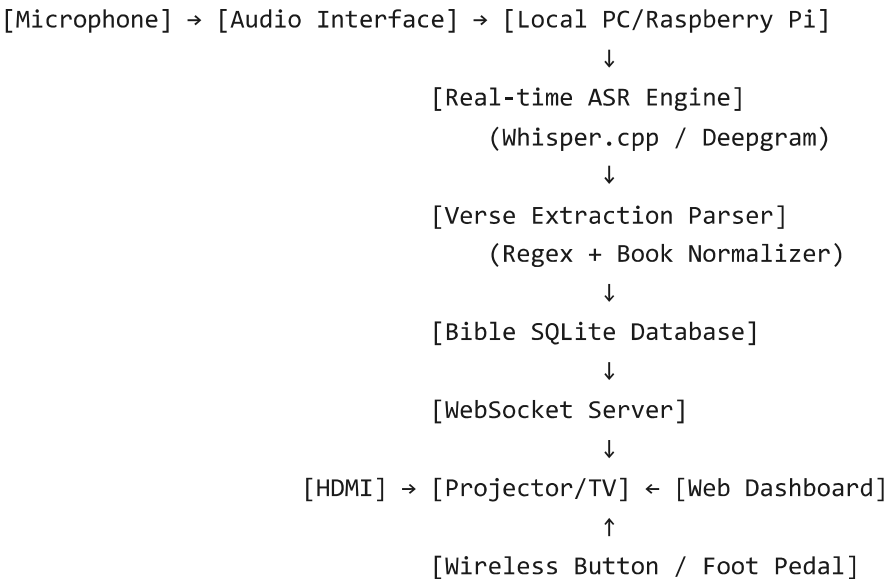
- Multiple mic array • Cloud dashboard (optional) • Sermon verse history + export • Mobile clear control

8. Out of Scope (MVP)

- Multi-language support (non-English Bibles)
- Voice commands beyond verse detection ("clear screen", "next verse")
- Automatic slide advance for worship software
- Mobile app for congregation (future companion app)

- Cloud dashboard across multiple churches

9. High-Level Architecture



10. Hardware Tiers (Budget)

Tier	Cost	Components
Basic	\$300–400	Raspberry Pi 4 (8GB), USB mic (Blue Snowball), wireless button
Standard	\$700–900	Intel NUC / mini PC, Shure MV7 mic, foot pedal, HDMI out
Pro	\$1400–1800	Dedicated fanless PC, wireless lapel mic, operator tablet, extra displays

 **Next Steps:** Build technical spike (offline Whisper on 10min of preaching audio) → source pilot church → finalize hardware bundle → develop MVP.

